

GPS Concepts

- measure distance to ≥ 4 satellites
- satellites transmit their positions in orbit
- user solves for position $P(x, y, z)$

GNSS Signals

- L band radio frequency
- ranging codes: range measurements
- navigation data: satellite coordinates
- Users can compute
 - range measurement: travelling time from satellite to receiver
 - also called pseudorange
 - satellite coordinates at any epoch
- Ranging codes
 - binary, 0 and 1
 - also called Pseudo Random Noise (PRN)
- Navigation Data
 - binary
 - satellite ephemeris information (Keplerian elements or position and velocity)
 - clock bias, almanac, health status
- Carrier
 - radio frequency, analog
 - sinusoidal signal

Example

Carrier L1: 1575.42 Hz

C/A PRN: 1.023 MHz

Chip duration: $\sim 1 \mu s$

C/A Period: 1023 chips

Navigation info: 50 Hz

Bit duration: 20 ms

One bit: 20 full PRN

GPS Modernization

- Signals: L1 C/A, L2C, L5
- L2C combines with L1 C/A
 - ionospheric correction
 - boosts accuracy
 - easier to receive (trees, indoors)
 - fast signal acquisition
- L5
 - reserved exclusively for aviation safety services
 - higher power, or greater bandwidth
- L1C
 - latest civil signal
 - same frequency as legacy L1 C/A
 - interoperability between GPS and international satellite navigation systems

$$f_0 = 10.23 \text{ MHz}$$

$$L1 = 154 f_0 = 1575.42 \text{ MHz}$$

$$L2 = 120 f_0 = 1227.60 \text{ MHz}$$

$$L5 = 115 f_0 = 1176.45 \text{ MHz}$$

Signal Polarization

- Linear and Circular
- linear: horizontal, vertical
 - TX $\neq$ RX, no signal
- circular polarized
 - work under any orientation
 - reduce multipath
 - remove distortion
 - clockwise: right hand anti-clockwise: LH

GPS Signal Generation

PRN Code generator

1.023 MHz binary

Data Generator

(C/A code)

50 bits/s binary
(Hz)

Carrier frequency generator

1575.42 MHz analog

+

mod-2 add / exclusive or

0 becomes 1

1 becomes -1

×

multiply

transmitted
satellite signal

- each carrier frequency is BPSK (binary phase shift keying) modulated by ranging codes
- ranging codes are different for different satellites, and are orthogonal to each other
- enables users to access signals from different satellites, same frequency without interference

L1 uses C/A code, L2 uses P code

(1.023 MHz)

(10.23 MHz)

In-phase (I) and quadrature (Q) component

- $I + 90^\circ = Q$
- I is sine wave, Q is cosine wave
- I/Q signals are amplitude modulated
 - efficient use of band width
 - 2 separate signals, same frequency band
 - they are orthogonal (do not interfere)
 - double data rate without additional bandwidth

L1 C/A Code

- C/A stands for coarse acquisition
- is pseudo random noise (PRN)
- accuracy is 1% chip length ($300m \times 1\% = 3m$)
- continually repeated every millisecond
- open to the public

GPS C/A Code (Gold code)

- PRN = 1 C/A code and the receiver generated same PRN C/A code have maximum correlation
- lag = 0, correlation value = 1023

GPS P Code (Precise)

- 10x chipping rate of C/A (10.23 MHz)
- chip distance = $c / \text{chip rate}$
- 10x accuracy of C/A (0.3 m)
- code sequence is extremely long 2.35×10^{14} chips (C/A has only 1023)
- repeats itself every 266 days
- each satellite is assigned a one week segment of P code
- in anti-spoofing mode, P code is encrypted into Y code

Almanac

- a navigation data
- enable computation of all the satellites in the constellation
- reduced accuracy, $\sigma = 1-2 \text{ km}$

GPS Navigation Data

- on L1, L2: 50 bps
- the whole message has 25 pages, 30s each

Signal Demodulation

- remove PRN = 1 C/A code
- the recovered carrier phase is the same
- auto-correlation

Signal Power Calculation

$$P_{dBW} = P_{dBm} - 30 \quad \text{power density } \left(\frac{dBW}{Hz} \right)$$
$$dBW = 10 \log_{10}(W) = 10 \log_{10} \left(\frac{\text{power (W)}}{\text{bandwidth (Hz)}} \right)$$
$$\text{spread/bandwidth} = 2 \cdot Hz$$

GPS Carrier Phase and Pseudorange Observable

$$\Phi = \rho + d\rho + c(dt - dT) - I + T + e_{m\phi} + \lambda N + n_{\phi}$$

$$P = \rho + d\rho + c(dt - dT) + I + T + e_{mp} + n_p$$

Φ : carrier phase range (m)

P : pseudorange code range (m)

$d\rho$: orbit error

dt : receiver clock error

dT : satellite clock error

I : ionospheric error T : tropospheric error

N : ambiguity in carrier phase range (cycle)

e : multipath error

n : noise

- error at broadcasting
 - satellite orbit, clock error
- error in propagation
 - ionospheric error, tropospheric error
- error at receiving
 - receiver noise, clock error
 - site multipath error

Identical Errors: orbit error, clock error (satellite)

Spatially correlated errors: ionospheric, tropospheric

6 Keplerian Elements

- a : semi-major axis
- e : eccentricity
- θ : true anomaly
- i : inclination
- Ω : right ascension (longitude) of the ascending node
- ω : argument of perigee

Keplerian Orbit

- force model
- inertial reference frame
- improve broadcast orbit accuracy

Satellite Clock Error

$$\Delta t = a_0 + a_1(t - t_{toc}) + a_2(t_r - t_{toc})$$

a_0 : clock offset

a_1 : clock drift

a_2 : clock drift rate

Total Electron Content (TEC)

- describe the ionosphere of earth
- no. of electrons between two points, along a tube of 1m^2 cross section

$$\text{TECU} = 10^{16} \text{ electrons/m}^2$$

$$\text{TEC} = 50 \text{ TECU} = 5 \times 10^{17} \text{ electrons/m}^2$$

- strongly affected by solar activity (11y solar cycle)
- measured by GPS 50×10^{16}

Multipath effects

- reflected signals
- radio signal reaches the receiver via 2 or more paths
- cannot be removed via differencing

Antenna Improvement

- Choke Ring Antenna
 - reject ground-bounce multipath
 - not useful for multipath from above
- Antenna cut-off pattern ($\geq 15^\circ$)
- left hand circular polarization (LHCP) rejection properties

- Suggestions:
- less weight for satellites with short period of tracking
 - use high elevation satellites
 - site selection
 - antenna, receiver selection

Earth Atmospheric Effect - Delay of Refraction

- total tropospheric delay $\sim 2.5\text{m}$
- Zenith Tropospheric Delay (ZTD)
- 90% dry delay, 10% wet delay
- dry delay is easier to model
- dry delay is more accurate, calculated from the model
- or "hydrostatic delay", "non-hydrostatic delay"

$$ZTD = N_h + N_{wet}$$

- correction to the slant direction
- the larger the elevation angle, the smaller the delay (m)

Combinations of GPS Observables

- at least 6 observations from a single GNSS satellite

$$\lambda_i \Phi_i^P = p + dp + \dots \quad p_{i,f}^P = p + dp + \dots$$

$$\lambda_2 \phi_i^P = \rho + d\rho + \dots \quad p_{i,f_2}^P = \rho + d\rho + \dots$$

$$\lambda_5 \phi_i^p = \dots$$

- errors affected by frequency: I , λN , n

Differential Method

- improve GNSS positioning accuracy
- try to cancel / mitigate errors
- single / double / triple difference
- L1 / L2 measurements

Single Difference

- 2 stations, 1 satellite, same time
- Cancel dT (satellite clock error)

Double Difference

- 2 stations, 2 satellites, same time, close to each other
- cancel d_p, I, T

Triple Difference

- 2 stations, 2 satellites, same time and place, close to each other, 2 measurements t_1, t_2
- cancel λN

Notation ρ_{ij}^{pq} — satellite
 $\rho_{ij}(t_i)$ — time
 — station

Linear Combination

- increase baseline length in GNSS Survey
- increase GNSS Positioning accuracy

1(A): wide-lane carrier phase LC

$$-i=1, j=-1$$

1(B): wide-lane carrier phase LC

$$\alpha = \frac{f_1}{f_1 - f_2} \quad \beta = -\frac{f_2}{f_1 - f_2}$$

2: wide-lane pseudorange LC

$$- \alpha = \frac{f_1}{f_1 + f_2} \quad \beta = \frac{f_2}{f_1 + f_2}$$

3: narrow lane carrier phase LC

$$- i = 1, j = 1$$

4: idiosphere-free carrier phase LC

$$-\alpha = \frac{f_1^2}{f_1^2 - f_2^2} \quad \beta = -\frac{f_2^2}{f_1^2 - f_2^2}$$

5: IF pseudorange LC

$$\alpha = \frac{f_1^2}{f_1^2 - f_2^2} \quad \beta = -\frac{f_2^2}{f_1^2 - f_2^2} \quad (\text{same as carrier phase})$$

6: Geometry-free LC

$$\alpha = 1 \quad \beta = -1$$

- different from other LC: $\alpha + \beta \neq 1$

$$\textcircled{1} \lambda_{NL} = \frac{1}{\frac{j}{\lambda_1} + \frac{j}{\lambda_2}} \quad (m)$$

$$\textcircled{2} \quad f = if_1 + jf_2 = \frac{c}{\lambda_{NL}} \quad (\text{Hz})$$

$$\textcircled{3} \quad \alpha = \frac{j\lambda_{NL}}{\lambda_1} \quad \beta = \frac{j\lambda_{NL}}{\lambda_2}$$

(4) $LC = \alpha L_1 + \beta L_2$

$$= \rho + \lambda_{NL} (N_1 + N_2) - \left(\frac{\lambda_{NL}}{\lambda_1} + \frac{\lambda_{NL}}{\lambda_2} \right) I, \quad 4$$

Inverse GPS

- a ground network to observe a single satellite
- 4 observations: x, y, z, dT (sat. clock error)
- same principle as GPS positioning
- need a global network to cover the earth

Basic dynamics - simple orbital motion assumptions:

- earth is a point mass
- mass of satellite is ignored
- no atmospheric drag or solar radiation pressure
- no gravitational attraction on the satellite
- Kepler's third law
 - $p^2 \propto a^3$
 - orbital period is proportional to size of orbit

- Period of the satellite

$$T = \frac{2\pi}{\sqrt{G \cdot M_{\text{earth}}}} a^{\frac{3}{2}} = \frac{2\pi}{\sqrt{\mu}} a^{\frac{3}{2}} \quad (\text{seconds})$$

$$G \cdot M_{\text{Earth}} = \mu = 3.986 \times 10^{14} \text{ (m}^3/\text{s}^2)$$

Orbital Calculation in Orbital Plane

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \cdot \cos \theta \\ r \cdot \sin \theta \\ 0 \end{bmatrix} \quad \begin{array}{l} r: \text{orbit radius} \\ \theta: \text{true anomaly} \end{array}$$

$$e = \sqrt{\frac{a^2 - b^2}{a^2}}$$

$$r = \sqrt{AS^2 + AF^2} = a \cdot (1 - e \cos E)$$

$$\text{mean motion } n = \sqrt{\frac{\mu}{a^3}}$$

GPS Reference Frame - WGS 84

- earth centered, earth fixed

Transformation of satellite coordinates in orbital plane to Earth-Centered-Inertial (ECI) coordinates

$$R_z(-\omega) = \begin{bmatrix} \cos \omega & -\sin \omega & 0 \\ \sin \omega & \cos \omega & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{1, \text{ECI}} = R_z(-\omega) \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{2, \text{ECI}} = R_x(-i) \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{1, \text{ECI}}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3, \text{ECI}} = R_z(-\Omega) \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{2, \text{ECI}}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{4, \text{ECI}} = R_z(-\Omega_0) \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3, \text{ECI}}$$

$\dot{\Omega}_0$ is the rate of ascension of ascending node

t_{oe} : epoch of ephemeris reference epoch

$$\Omega_0 = (\dot{\Omega} - \omega_e) \cdot t_k - \omega_e \cdot t_{oe}$$

Validity of GPS Ephemerides

- ephemerides are valid up to 4 hours
- GPS control segment updates ephemerides every 2 hours
- GPS almanac is updated weekly through 3 ground-based control stations
- GPS almanac is valid up to months

Key Questions

- What is Pseudorange?
- Why do we use Pseudorange?
- How to use Pseudorange?

$$\frac{f_1^2 \cdot f_2^2}{\frac{1}{f_1^2} - \frac{1}{f_2^2}} = \frac{P_2 - P_1}{40.3 \cdot 10^{16}}$$

$$\frac{P_1 - P_2}{40.3 \cdot (1-\gamma)} = f_1^2 \text{ where } \gamma = \frac{f_1^2}{f_2^2}$$

$$\text{TEC} = \int$$

$$P_1 - P_2 = p + c(dt - [\Delta t_{sv} - T_{gd}]) + T + \frac{40.3 \text{ TEC}}{f_1^2}$$

$$= \{ p + c(dt - [\Delta t_{sv} - T_{gd}]) + T + \frac{40.3 \text{ TEC}}{f_2^2} \}$$

$$= (1-\gamma) T_{gd} + (1-\gamma) \frac{40.3 \text{ TEC}}{f_1^2}$$

yields

$$\rightarrow f_1^2 \cdot \frac{(P_1 - P_2) - (1-\gamma) T_{gd}}{40.3(1-\gamma)} \quad \gamma = \frac{f_1^2}{f_2^2}$$

$$\gamma \frac{40.3 \text{ TEC}}{f_1^2} = \frac{40.3 \text{ TEC}}{f_2^2}$$

Positioning with Carrier Phase

$$\lambda \phi_i^P = \rho_i^P + d\rho_i^P + c(\delta t_i - \delta T^P) - I_i^P + T_i^P + \lambda N_i^P + \sigma_i^P$$

- e.g. 3 satellites $\rightarrow$ 2 double differencing
2-1, 3-1

- potentially mm accuracy

Typical Algorithm: Double Difference

- cancels:
 - $c \nabla \delta T^{pq}$
 - $\Delta \nabla \text{orb}_{ij}^{pq}$
 - $\Delta \nabla \text{lon}_{ij}^{pq}$
 - $\Delta \nabla \text{Trop}_{ij}^{pq}$
- still present: $\lambda \Delta \nabla N_{ij}^{pq}$, $\Delta \nabla \sigma_{ij}^{pq}$
- require short baseline

Errors in double differencing

- ionospheric scintillation
- Solid Earth tide
 - geodetic networks and PPP must include
- atmospheric tide
 - oscillations of atmosphere
 - heating by solar radiation
 - Moon gravity
 - planetary waves, convections
 - 2 daily peaks and valleys

Linear Model

- if s satellites tracked for an epoch:
 - Independent observation equations: $s-1$
 - for k epochs $k(s-1)$
 - unknown parameters: $s+2$
 - $(s-1)$ DD ambiguities
 - 3 coordinates

$$\hat{L} = B\hat{b} + A\hat{a} = [B \ A] \begin{bmatrix} \hat{b} \\ \hat{a} \end{bmatrix}$$

Variance - Covariance Matrix

$$\Delta \nabla \hat{\phi}_{ij}^{12} = (\hat{\phi}_i^1 - \hat{\phi}_i^2) - (\hat{\phi}_j^1 - \hat{\phi}_j^2)$$

...

- error propagation law

$$\begin{bmatrix} \Delta \nabla \hat{\phi}_{ij}^{12} \\ \Delta \nabla \hat{\phi}_{ij}^{13} \\ \vdots \end{bmatrix}_{l+v} = [A] \begin{bmatrix} \hat{\phi}_i^1 \\ \hat{\phi}_i^2 \\ \hat{\phi}_i^3 \\ \vdots \\ \hat{\phi}_j^1 \\ \hat{\phi}_j^2 \\ \hat{\phi}_j^3 \\ \vdots \end{bmatrix}_x$$

$$\begin{bmatrix} \sigma_{\Delta \nabla \hat{\phi}_{ij}^{12}}^2 & \sigma_{\Delta \nabla \hat{\phi}_{ij}^{12} \Delta \nabla \hat{\phi}_{ij}^{13}} \\ \sigma_{\Delta \nabla \hat{\phi}_{ij}^{13} \Delta \nabla \hat{\phi}_{ij}^{12}} & \sigma_{\Delta \nabla \hat{\phi}_{ij}^{13}}^2 \\ \vdots & \vdots \end{bmatrix} = A^T P A$$

varcov

- more than 3, varcov looks like $\sigma_{\hat{\phi}}^2 \begin{bmatrix} 4 & \dots & 2 \\ \vdots & \ddots & \vdots \\ 2 & \dots & 4 \end{bmatrix}$
- assume independent
- in reality GNSS errors are correlated
 - obs in same epoch (e.g. L1, L2)
 - location (e.g. T)

LS Solution

- float DD solution
- estimated ambiguities are float, not int

Ambiguity fixed solution

- try to solve DD ambiguities (a) to int (a^*)
- a^* is all/partial int vector

$$(I - Aa^*) + v = Bb^*$$

$\rightarrow b^*$ is DD fixed solution

- disadvantage: slow resolution

- r^d (e.g. search range=11, DD observable=19
 $\rightarrow 11^{19}$)

Hatch's method (LSAST)

- 8 cubic points
- point is in the cube (float ambiguity)
- unknowns: only 3 coordinates
→ no. of satellites: 4
- normally get Δ ambiguity, then position
- Hatch's method: position first, then ambiguity
- 8 candidates (8 corner points)
- 3 equations, 3 ambiguities
eg. $\nabla \Delta \phi^{',10} = \nabla \Delta \rho^{',10} + \lambda \nabla \Delta N^{',10}$
 $\nabla \Delta \phi^{',15} = \nabla \Delta \rho^{',15} + \lambda \nabla \Delta N^{',15}$
...
- 8 candidates → 8 sets of 3 ambiguities
- no. of searches = $[\text{max ambiguity} - \text{min ambiguity}] + 1$
- 1 set = 1 position = +1 to total #

Basic Procedure of Data Processing

① Preprocessing

- smoothing pseudorange using carrier phase data
- approximate coordinates from pseudorange
- editing cycle slips (i.e., cutoff angle $\geq 15^\circ$)

② Least Squares Estimator to solve for all parameters (float solution)

- float ambiguity resolution
- float solution (station coordinates with float ambiguities)

③ Fix Ambiguity

- ambiguity resolution method

④ → determine b^*

Ambiguity-fixed solution

- input fixed ambiguity as known values
- solve for stn coordinates again → Fixed Solution

Precise Point Positioning (PPP)

- purpose:
 - airborne mapping
 - large coverage / long distance
 - RTK accuracy not enough / RTK nil
 - offshore (i.e. rigs)
 - deserts

PPP vs DGPS / RTK

- PPP is cheaper, more convenient logistically

Advantage of PPP

- single receiver operation
- cost saving (equipment and labor)
- use global reference frame
- simple logistics in the field
- absolute value of some products (water vapor, TEC)
- high accuracy (cm/dm)
- Requires:
 - precise $d\rho$, δT , λN (fractional) correction

GPS Point Positioning

- with 1 GPS receiver
- two categories:- Standard Positioning Service (SPS)

SPS vs PPP

- | | |
|-----------------------------------|--|
| - 1 single frequency Rx | - 1 dual frequency Rx |
| - GPS broadcast orbit, clock | - IGS precise orbit, clock |
| - pseudorange | - carrier phase |
| - P does not consider λN | - λN must be resolved as float/fixed/partially float/fixed solution |
| - m level | - cm-dm level |
| - obtained instantly | - not instantaneous |

Antenna Phase Center Error

- sent / received point $\neq$ measuring point
- mitigation: use IGS atx

PPP Biases

- Hardware, firmware biases
 - sv, rx
- Code measurements
 - sv, rx
- Phase measurements
 - sv, rx

$$L_1 = \rho + d_{orb} + c(dt - dT) - I_1 + T - \lambda N_1 + b_{L_1}^r - b_{L_1}^s + \varepsilon_{L_1}$$

$$L_2 = \dots - I_2 \dots - \lambda N_2 \dots$$

Ionosphere-free Combination Observation

$$L_3 = \frac{f_1^2}{f_1^2 - f_2^2} L_1 - \frac{f_2^2}{f_1^2 - f_2^2} L_2 \quad (\alpha L_1 + \beta L_2)$$

$$= \rho + d_{orb} + c(dt - dT) + T - \lambda_3 N_3 + \varepsilon_{L_3} +$$

$$\left(\frac{f_1^2}{f_1^2 - f_2^2} b_{L_1}^r - \frac{f_2^2}{f_1^2 - f_2^2} b_{L_2}^r \right) - \left(\frac{f_1^2}{f_1^2 - f_2^2} b_{L_1}^s - \frac{f_2^2}{f_1^2 - f_2^2} b_{L_2}^s \right)$$

(original I)

$$= \rho + d_{orb} + c(dt - dT) + T - \lambda_3 N_3 + \varepsilon_{L_3} + \underline{b_{L_3}^r - b_{L_3}^s}$$

$$f_0 = 10.23 \text{ MHz}$$

$$f_1 = 154 \times f_0 = 1575.42 \text{ MHz}$$

$$f_2 = 120 \times f_0 = 1227.6 \text{ MHz}$$

For wide lane:

carrier: $\alpha = \frac{f_1}{f_1 - f_2}, \beta = -\frac{f_2}{f_1 - f_2}$

pseudorange: $\alpha = \frac{f_1}{f_1 + f_2}, \beta = +\frac{f_2}{f_1 + f_2}$

For narrow lane:

carrier: $i = 1, j = 1$

pseudorange: $\alpha = \frac{f_1}{f_1 + f_2}, \beta = +\frac{f_2}{f_1 + f_2}$
(same as wide lane)

For ionofree P/L:

$$\alpha = \frac{f_1^2}{f_1^2 - f_2^2}, \beta = -\frac{f_2^2}{f_1^2 - f_2^2}$$

(wide lane, but f_1^2, f_2^2)

Melbourne-Wübbena combination observation

$$L_3 = \rho + d_{orb} + c(t_{L_3}^r - t_{L_3}^s) + T - \lambda_3 N_3 + \varepsilon_{L_3}$$

$$= \rho + d_{orb} + c(t_{L_3}^r - t_{L_3}^s) + T - \lambda_3 (17N_1 + 60N_4) + \varepsilon_{L_3}$$

$$A_4 = (b_{A_4}^r - b_{A_4}^s) - \lambda_4 N_4 + \varepsilon_{A_4} \quad /$$

- int ambiguity

Derivation of A_4

$$A_4 = L_4 - P_6$$

$$= -\lambda_4 N_4 + \left[\frac{f_1}{f_1 - f_2} b_{L_1}^r - \frac{f_2}{f_1 - f_2} b_{L_2}^r - \left(\frac{f_1}{f_1 + f_2} b_{P_1}^r + \frac{f_2}{f_1 + f_2} b_{P_2}^r \right) \right]$$

$$- \left[\frac{f_1}{f_1 - f_2} b_{L_1}^s - \frac{f_2}{f_1 - f_2} b_{L_2}^s - \left(\frac{f_1}{f_1 + f_2} b_{P_1}^s + \frac{f_2}{f_1 + f_2} b_{P_2}^s \right) \right]$$

$$+ \varepsilon_{A_4}$$

let $\alpha_L = \frac{f_1}{f_1 - f_2}, \beta_L = -\frac{f_2}{f_1 - f_2}$

$\alpha_P = \frac{f_1}{f_1 + f_2}, \beta_P = +\frac{f_2}{f_1 + f_2}$

$$= -\lambda_4 N_4 + \left[\alpha_L b_{L_1}^r + \beta_L b_{L_2}^r - (\alpha_L b_{P_1}^r + \beta_L b_{P_2}^r) \right]$$

$$- \left[\alpha_L b_{L_1}^s + \beta_L b_{L_2}^s - (\alpha_L b_{P_1}^s + \beta_L b_{P_2}^s) \right]$$

$$+ \varepsilon_{A_4}$$

$$= -\lambda_4 N_4 + (b_{A_4}^r - b_{A_4}^s) + \varepsilon_{A_4} //$$

Decoupled clock model

- N_1, N_4 : int
- for each sv, Rx: 3 corrections

sv decoupled code clock	Rx code clock
phase clock	phase clock
sv A4 bias	A4 bias
- 3 equations for each satellite
 - iono code obs
 - iono phase obs
 - MW combination obs

PPP Unknowns

for N satellites: $3N$ observations

total unknowns: 3 coordinates of Rx

1 tropospheric parameter

3 Rx decoupled clock parameters

$3N$ Sv decoupled clock parameters

$2N$ ambiguities (N_1, N_4 for each satellite)

↳ $5N+7$ unknowns

- therefore cannot directly use least square

Network solution

- multiple PPP receivers (M receivers)

- total observations: $3MN$ (double counted = $3N(M-1)$)

- pseudo observations: $2N+2M+1$

- total: $3MN+2N+2M+1$

- number of unknowns: $(5N+7) \times M$

- actual unknown = unknown - double counted

$$= (5N+7) \times M - 3N(M-1)$$

- no. of redundancy = all observations - actual unknown

$$= (3MN+2N+2M+1) - [(5N+7) \times M - 3N(M-1)]$$

- require ≥ 7 tracked satellites